WORKED SOLUTION

STEP II 2012, Question 5

Pure • Curve sketching

STEP II 2012, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q5

The key idea

Both curves are reciprocals, so we sketch the denominator in our heads and flip it: where the denominator is zero we get a vertical asymptote, where it is large y is near 0, and a turning point of the denominator is a turning point of the curve. In part (ii) the two quadratic factors are mirror images of each other about the midpoint of a and b, so shifting the origin to that midpoint makes g an even function of the new variable and the stationary points fall out of a quadratic in t^2.

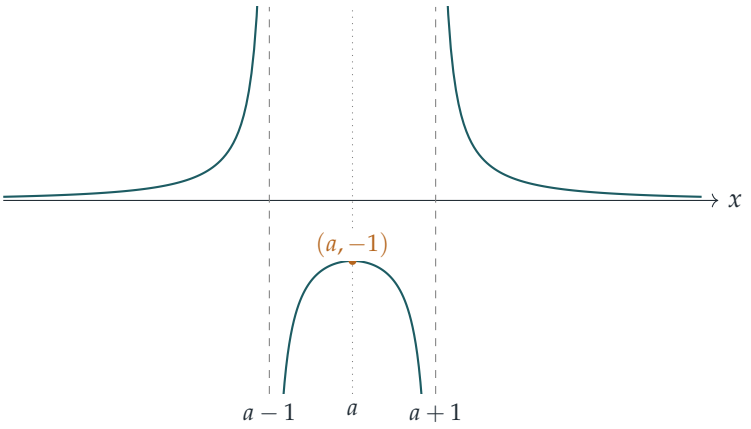
Solution

Part (i)

Write $D(x) = (x-a)^2 - 1$, so $f = 1/D$. The denominator D is a parabola, symmetric about $x = a$, with minimum value -1 at $x = a$ and zeros at $x = a \pm 1$. Reading off the reciprocal:

- The curve is symmetric about $x = a$, with vertical asymptotes $x = a \pm 1$.
- For $|x-a| < 1$ we have $-1 \leq D < 0$, so $f \leq -1$, with equality only at $x = a$. So $(a, -1)$ is a local maximum, and $f \to -\infty$ as x approaches $a \pm 1$ from inside.
- For $|x-a| > 1$ we have $D > 0$ and D increases as $|x-a|$ grows, so f is positive and decreases from $+\infty$ (next to the asymptotes) towards 0. The x-axis is a horizontal asymptote, with $f \approx 1/x^2$ for large $|x|$.

Indeed $f'(x) = -2(x-a)/D^2$ vanishes only at $x = a$. The curve crosses the y-axis at $1/(a^2-1)$ (when $a \neq \pm 1$), but where that lies depends on a, so we draw the picture relative to $x = a$.



Part (ii)

Let $p = (x-a)^2 - 1$ and $q = (x-b)^2 - 1$, so $g = 1/(pq)$. Put

$$c = \tfrac{1}{2}(a+b), \qquad d = \tfrac{1}{2}(b-a) > 0, \qquad t = x - c,$$

so that $x - a = t + d$ and $x - b = t - d$. Then, using $(t+d)(t-d) = t^2 - d^2$ and $(t+d)^2 + (t-d)^2 = 2t^2 + 2d^2$,

$$\begin{aligned} pq &= \big((t+d)^2 - 1\big)\big((t-d)^2 - 1\big) \\ &= (t^2 - d^2)^2 - (2t^2 + 2d^2) + 1. \end{aligned}$$

This depends only on t^2, so g is symmetric about $x = c$, the midpoint of a and b. Writing $u = t^2$ and $h(u) = (u - d^2)^2 - 2u - 2d^2 + 1$,

$$\frac{\mathrm{d}(pq)}{\mathrm{d}t} = 2t\,h'(u) = 2t\big(2(u - d^2) - 2\big) = 4t\,(t^2 - d^2 - 1).$$

Since $g' = -(pq)'/(pq)^2$, the stationary points of g are where $(pq)' = 0$:

$$t = 0 \qquad \text{or} \qquad t = \pm\sqrt{d^2 + 1},$$

provided these are not asymptotes. The asymptotes are at $t = \pm(d+1)$ and $t = \pm(d-1)$, and since $d < \sqrt{d^2+1} < d+1$, the outer two stationary points are always genuine. In terms of a and b the candidates are

$$x = \frac{a+b}{2} \qquad \text{and} \qquad x = \frac{a + b \pm \sqrt{(b-a)^2 + 4}}{2}.$$

The values there are $h(0) = (d^2-1)^2$ and $h(d^2+1) = 1 - 2(d^2+1) - 2d^2 + 1 = -4d^2$.

Ask yourself Which of these candidates actually survive, and are they maxima or minima?

It depends on whether $t = 0$ coincides with an asymptote, that is, whether $d = 1$; this is exactly the split between $b > a + 2$ (so $d > 1$) and $b = a + 2$ (so $d = 1$). For the nature of each point, we look at the sign of g on each interval between asymptotes and its behaviour at the ends, rather than differentiating again.

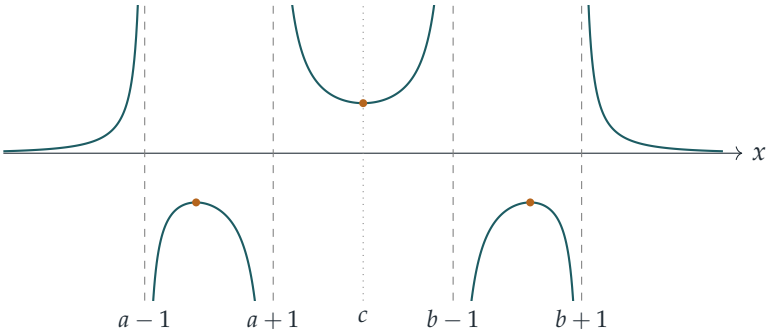
Case $b > a+2$. Now $d > 1$, and the four asymptotes, in order, are

$$t = -(d+1),\ -(d-1),\ d-1,\ d+1, \quad \text{i.e.} \quad x = a-1,\ a+1,\ b-1,\ b+1,$$

all distinct, with c strictly between $a+1$ and $b-1$.

- *Outside, $|t| > d+1$:* here $|x-a| > 1$ and $|x-b| > 1$, so $p, q > 0$ and $g > 0$. There are no stationary points, so g falls from $+\infty$ at the asymptote to 0 at infinity, behaving like $1/x^4$.
- *Middle, $|t| < d-1$, i.e. $a+1 < x < b-1$:* again $p, q > 0$, so $g > 0$, and $g \to +\infty$ at both ends. The only stationary point is $t = 0$, which must therefore be a minimum, with value
$$g(c) = \frac{1}{(d^2-1)^2} = \frac{16}{\big((b-a)^2-4\big)^2}.$$
- *The two gaps, $d-1 < |t| < d+1$:* for t in $(d-1, d+1)$, we have $|x-b| < 1$ but $x - a > 2d - 1 > 1$, so $q < 0 < p$ and $g < 0$, with $g \to -\infty$ at both ends. The only stationary point there is $t = \sqrt{d^2+1}$, which must be a maximum, with value
$$g = \frac{1}{-4d^2} = -\frac{1}{(b-a)^2}.$$
The gap $-(d+1) < t < -(d-1)$ is the mirror image.

So there are three stationary points: a local minimum at $x = \frac{1}{2}(a+b)$ and two local maxima, both with $y = -1/(b-a)^2$, at $x = \frac{1}{2}\big(a+b \pm \sqrt{(b-a)^2+4}\big)$. The sketch below has $b - a = 4.8$, with the y-direction stretched so that the turning points show.



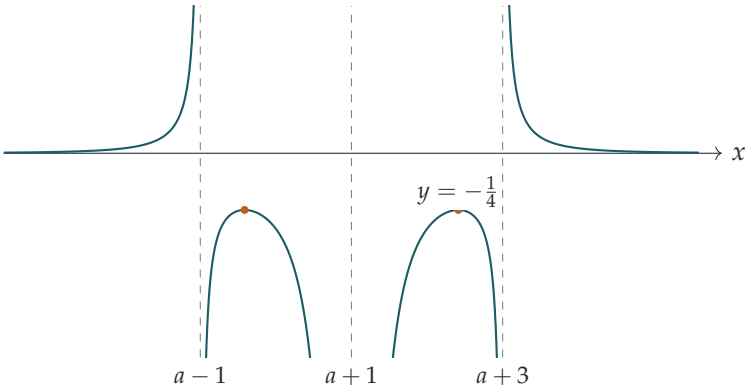
Case $b = a+2$. Now $d = 1$, so $a+1 = b-1 = c$ and the two middle asymptotes merge into one, at $x = a+1$. With $t = x - a - 1$,

$$pq = h(t^2) = t^4 - 4t^2 = t^2(t^2-4), \qquad g = \frac{1}{t^2(t^2-4)},$$

so there are three asymptotes, $x = a-1,\ a+1,\ a+3$. The candidate $t = 0$ is now an asymptote, not a stationary point, which leaves only $t = \pm\sqrt{2}$.

- For $|t| > 2$, $g > 0$ and g falls from $+\infty$ to 0, as before.
- For $0 < |t| < 2$, $g < 0$, and $g \to -\infty$ both as $t \to 0$ (where $g \approx -1/(4t^2)$, so both sides go down) and as $|t| \to 2$. The stationary point at $t = \sqrt{2}$ is therefore a maximum, with $g = 1/\big(2 \cdot (-2)\big) = -\frac{1}{4}$; likewise at $t = -\sqrt{2}$.

So the stationary points are at $x = a + 1 \pm \sqrt{2}$, both local maxima with $y = -\frac{1}{4}$. (This agrees with the general formula: with $b - a = 2$ it gives $x = \frac{1}{2}(2a + 2 \pm \sqrt{8})$, and $-1/(b-a)^2 = -\frac{1}{4}$.) The curve now never goes above the x-axis between $x = a - 1$ and $x = a + 3$.



Checks

- With $a = 0$, $b = 5$: $g(2.5) = 1/\big((6.25 - 1)^2\big) = 1/27.5625$, and the formula $16/(25 - 4)^2 = 16/441$ gives the same, about 0.0363. At $x = \frac{1}{2}(5 + \sqrt{29}) \approx 5.193$, direct evaluation gives $g \approx -0.04 = -1/25$.
- Letting $b \to a + 2$ in the first case, the minimum value $16/\big((b-a)^2 - 4\big)^2$ blows up, matching the middle branch disappearing into the merged asymptote.
- A numerical scan of the sign changes of g' for random a and b finds exactly three stationary points when $b > a + 2$ and two when $b = a + 2$, at the positions above.

Common slips

- Expanding pq as a quartic in x and trying to factorise the cubic $(pq)'$ directly: it works, but the symmetric substitution $t = x - c$ makes the factorisation immediate.
- Forgetting that in the case $b = a + 2$ the point $x = a + 1$ is an asymptote (a double zero of the denominator), not a turning point.
- Getting the sign of g wrong in the gaps: exactly one of p, q is negative there, so $g < 0$.

Beyond STEP

Sketching $1/D$ from D is a first taste of thinking about functions qualitatively: sign, zeros, symmetry and behaviour at infinity tell you almost everything before any calculus. The shift $t = x - \frac{1}{2}(a+b)$ is a use of symmetry to reduce a quartic to a quadratic in t^2, a trick that recurs throughout algebra and analysis. For the careful treatment of limits and asymptotes behind such sketches, see Abbott, *Understanding Analysis*, or for many more problems of this kind, Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.